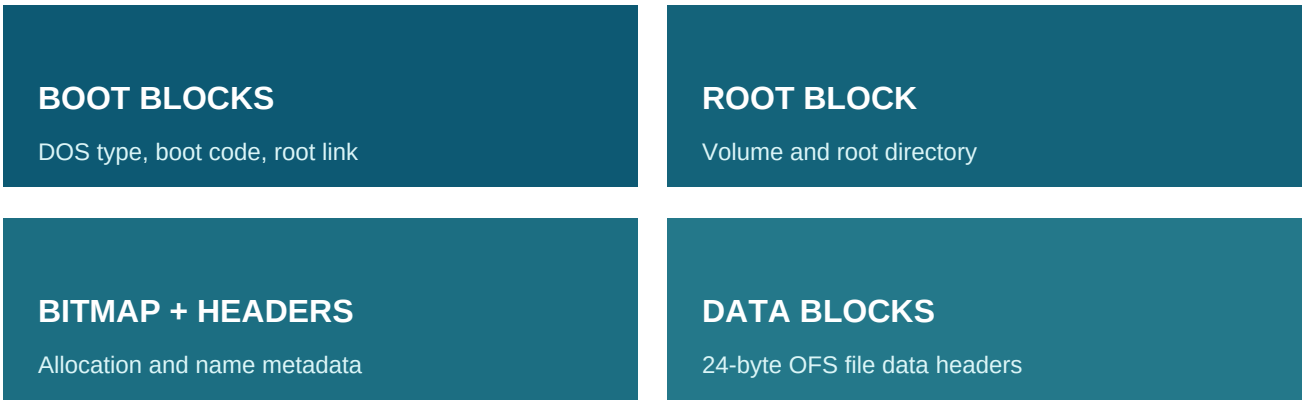
Amiga OFS - original disk filesystem

ORIGINAL FORMAT / HISTORY / ENGINEERING DESIGN

Amiga Original File System (DOS0) preserves checksummed header-bearing data blocks, a bitmap allocator, and AmigaDOS name and directory semantics.

This reference describes the original upstream disk format and its historical engineering rationale. The separate InfiltratorOS adapter is not the original specification. Historical facts, format rules and implementation limitations are separated; consult the named primary sources for field-level normative requirements.

ON-DISK STRUCTURE / CONCEPTUAL MAP



CHAPTERS

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HISTORICAL CHRONOLOGY

The original Amiga file system appeared with the first Commodore Amiga machines in 1985. The DOS0 format suited the floppy-centred Amiga environment and its operating system's distinctive handler and device model. The Fast File System later changed file-data layout for better throughput on hard disks; it was not merely an optimised implementation of identical OFS data blocks.

Amiga filesystem modes carry DOS type identifiers: original DOS0, international DOS2, directory-cache DOS4 and later long-name variants. The AmigaOS documentation identifies the international change with correct handling of accented filename case equivalence, while directory caching was introduced to accelerate floppy directory listing. These variants are historical format decisions that require explicit handling, not silent assumptions about every Amiga disk.

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ORIGINAL DESIGN AND ON-DISK STRUCTURES

OFS data blocks include a 24-byte header with block type, ownership, sequence and checksum-related fields before the payload. File header and extension blocks carry pointers to data blocks, timestamps, names and comments. Root and directory header blocks maintain directory hash-chain links. Free and used blocks are tracked in on-media bitmap blocks, not in a FAT-style next-cluster table.

Amiga filesystems use block keys and structured blocks with checksums. The DOS type in the boot block identifies variant features. Header tables can point to data blocks in an order different from physical placement; reverse-oriented pointer tables and extension blocks must be interpreted according to the upstream format. Logical payload size is not simply equal to the number of allocated blocks times the block size, because OFS data block headers consume space.

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ALLOCATION AND DIRECTORIES

The root block and directory headers act as directories containing hashed link chains to child objects. Names, Amiga protection bits, comments and datestamps belong to on-disk entry metadata. File extension blocks are required when the direct header’s pointer capacity is exceeded. Different DOS variants add changes to the name comparison, directory cache and long-name representation.

Bitmap state identifies free blocks, but consistency also requires traversing every rooted object and tracing every referenced header, extension and data block. Filesystems may contain hard and symbolic link concepts that differ from Unix, and modern Linux adapters must map those semantics without inventing guarantees that the original format does not provide. Special care is needed when links form cycles.

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INTEGRITY, VALIDATION AND RECOVERY

OFS protects its structural and file data blocks with on-media checksums, unlike FFS data blocks. The checksums detect many forms of corruption but do not supply modern transactional rollback. The volume may also have bitmap-valid state and must not be declared writable after an interrupted update without checking allocation ownership and structural reachability.

Safe recovery requires checking root and directory structures, bitmap ownership, extension chains, data-sequence numbers, header checksums and links. A bitmap rebuild is acceptable only if the file tree provides an internally consistent ownership inventory. Failure during metadata changes can leave seemingly valid blocks with inconsistent higher-level ownership; inventing a missing parent risks permanent damage.

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DESIGN TRADE-OFFS AND HISTORICAL LIMITATIONS

OFS reflects floppy-disk priorities: inexpensive integrity checks, native Amiga metadata and explicit block ownership. Embedding 24-byte metadata headers in data blocks reduces useful data per block and creates additional work during sequential reads. FFS addressed part of this performance limitation by storing file data without OFS-style headers.

Old AmigaDOS formats are historically significant but are not uniform across DOS0/2/4/6. The directory-cache variation speeds some lookups at additional maintenance cost. International and long-name variants need compatible name comparison and encoding policies. Rigid assumptions about 512-byte blocks must not be used where larger blocks are permitted by the format.

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INFILTRATOROS INTERPRETATION

Our OFS.so offers Amiga-format operations through a versioned portable ABI. It is not the historical AmigaOS filesystem handler, and the separate optional Linux kernel adapter and EFI boot-loading mechanism are compatibility layers. The implementation’s current write/defragmentation claims must be checked against evidence in its own repository, not deduced from the original OS reference.

For exact DOS0, DOS2, DOS4 and DOS6 support, preserve the correct data-block headers, checksums, name rules, hash chains, link rules and bitmap protocols. Do not conflate OFS with Amiga Fast File System or use FFS file payload assumptions.

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PRIMARY SOURCES & DOCUMENTATION

References are included to make the historical and technical account auditable. The source documents, published format specifications and implementation references take precedence over this explanatory guide where a detail differs.

1. Linux kernel Overview of Amiga Filesystems (DOS0 through DOS5)

https://kernel.org/doc/html/v6.7/filesystems/affs.html

2. AmigaOS technical documentation, directory cache and long-name low-level structures

https://wiki.amigaos.net/wiki/DCFS_and_LNFS_Low_Level_Data_Structures

3. AmigaOS file system capability and compatibility table

https://wiki.amigaos.net/wiki/UserDoc%3AAmigaOS_File_Systems

Implementation boundary: the installed InfiltratorOS .so, the optional Linux kernel .ko, and the source-first installer are distinct products. The original filesystem design is documented above; claims about the adapter’s qualification require its own source, tests and release evidence.

OFS / technical detail

ON-DISK FIELD AND STRUCTURE INVENTORY

Format-specific field names and meanings are summarised below; where applicable the original specification defines exact endianness, offsets, checksums and format-revision limits.

1. Boot identifier: DOS0, DOS2, DOS4 or DOS6 variant identity.
2. Root block: volume name, time fields, bitmap and hash table.
3. Directory header: metadata and hash-chain continuation links.
4. File header: size, name, protection, extension and data-table pointers.
5. File extension: continuation of reversed data pointer sequence.
6. OFS data block: 24-byte type, owner, sequence, size, next, checksum header.
7. Bitmap blocks: per-block free/used ownership state.
8. Bitmap-valid protocol: distinguishes clean allocation bookkeeping.

OFS / technical detail

HISTORICAL FORMAT DEVELOPMENT

Selected design milestones explain why this format acquired its present layout. Dates refer to historical format or implementation changes, not InfiltratorOS releases.

1. 1985: original AmigaDOS OFS shipping with the first Amiga.

2. Late 1980s: FFS removes file data block headers for speed.

3. 1992: directory-caching file system mode introduced.

4. 2001: long-name variation extends AmigaDOS structures.

Important historical distinction: newly introduced driver features are not necessarily changes to the original on-disk format. Read the version or feature bits rather than assuming current software support defines historic media.

OFS / technical detail

FORMAT VALIDATION AND RECOVERY INVARIANTS

A writable implementation must establish all of these conditions before modifying any original-format media. These checks describe the minimum class of invariants, not proof of complete recovery qualification.

1. Check 24-byte OFS data header and checksum for each file block.

2. Require data sequences and owner keys consistent with file headers.

3. Validate bitmap allocation against namespace reachability.

4. Distinguish DOS0, international, cache and long-name modes.

5. Never apply FFS raw-data mapping to OFS blocks.

Forensic principle: use a separate verified image, retain original structures, and report conflicting evidence instead of inventing a new supposedly clean filesystem state.

OFS / engineering detail

DISK READ PATH / ENGINEERING NOTES

1. Confirm the DOS type in the boot blocks before interpreting a volume; DOS0, DOS2, DOS4 and DOS6 imply related but not identical filename and directory-cache behaviour.

2. Use Amiga big-endian fields and the partition-relative block numbering expected by the original format. Raw disk offsets must account for the partition's geometry and reserved blocks.

3. Locate and validate the root block and its directory hash table. The root is not a FAT-style fixed table at the start of the data region.

4. Resolve directory hash buckets through on-disk hash-chain links. Check the type and checksum of every header and prevent cycles in damaged directory chains.

5. Read file header and extension blocks to reconstruct ordered data-block references. Pointer table order must be interpreted correctly, not presumed to match physical block order.

6. An OFS data block carries a 24-byte header for block identity, file ownership, sequence, data length, successor and checksum information before its payload.

7. Compute useful payload size from the format's block and header layout. Allocated block count alone does not equal logical file length.

8. Verify bitmap references and free/allocated ownership separately from reachability; a set bit cannot prove that a block belongs to the claimed file.

OFS / engineering detail

VALIDATION, RECOVERY AND COMPATIBILITY

1. Verify checksum coverage using AmigaDOS block rules and the correct block length. A generic CRC algorithm is not equivalent to the format’s checksum.

2. Check that every file-data sequence advances consistently and belongs to the expected file header; blindly following a next pointer can misassociate unrelated data.

3. Preserve case-folding and accented-character behaviour appropriate to each international DOS variant. Do not force modern Unicode collation onto legacy name matching.

4. Directory-cache variants introduce additional metadata that must be updated coherently. A plain DOS0 writer must not silently modify a DOS4 volume.

5. Do not infer that every block size accepted by an adapter is valid for every historic medium or geometry. Read the volume definition first.

6. Distinguish data-block damage from header-chain damage; OFS checksum protection aids detection but is not a transaction journal.

7. On suspected corruption, inventory root-reachable blocks and bitmap ownership before rebuilding free-space metadata, and retain original images for independent recovery.

8. The original AmigaDOS format rules govern disk layout; InfiltratorOS .so and .ko support must be verified independently by tests.